Device for Chemical Demagnetization of Red Beds[1]

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Chemical demagnetization of red sandstones by means of high pressure HCl is an effective way of removing secondary magnetic components acquired in the modern weathering environments.

Cambro-Ordovician red sandstone samples from Wadi Ram, and Petra (29.7°N, 35.3°E), southern Jordanian desert show NRM (Natural Remanent Magnetization) directions that quite accurately reflect the present geomagnetic field. A magnetic pole calculated from these NRM directions would be situated at 160°E, 87°N, (N = 347, K = 36, α = 3.6) and its α_{95}-circle of confidence almost covers the positions of the pole of the present geocentric dipole (the virtual geomagnetic pole) and the presently observed geomagnetic pole. These NRM directions suggest that these Lower Paleozoic red beds, exposed to arid weathering conditions, have recently acquired a strong secondary magnetization.

Attempts to remove these secondary components by high- and low-temperature and alternating-field demagnetization techniques were unsuccessful for those intensely red-colored samples which were collected close to the modern weathering surface. Only samples from a few inches below the case hardening surface sometimes responded favorably to standard demagnetization methods.

High coercive forces (>2800 oe) and high Curie temperatures (>630°C) suggest that hematite is responsible for the rather stable secondary magnetic component. It is suspected that groundwater percolating through the sandstones continuously transports iron to the surface where Fe-hydroxides are precipitated, accounting for the red staining. Due to the arid conditions of the area, the Fe-hydroxides are dehydrated to hematite. This being the case, one eventually could approach the original DRM by removing the red staining causing the secondary CRM component [*Collinson*, 1967].

In treating red sandstones with HCl it was found that the sandstones do not disintegrate but are bleached without significant loss of cohesion. Usually, however, the central part of the standstone samples could not be adequately cleaned by simply immersing them in acid. Moreover, this bleaching process required several days of immersion, and the magnetic fraction of the heavy minerals was attacked as well. Hence a quick method was needed to bleach the samples without affecting the heavy minerals unduly.

Forcing HCl through the samples by means of compressed air meets these requirements. Experimentally, HCl with 3 N dilution was found to be the most efficient concentration and acid. Figure 1 shows the chemical demagnetization device. Essentially it consists of a chamber in which the sample can be sealed with 2 O-rings on both sides, an acid chamber that can hold about 500cc of HCl, and a connection to a regulated high pressure air supply. The acid is pressed through the inner upper O-ring and spreads within the sample. The Fe-chlorides and the HCl, then drip into a water bath below the device. Distilled water is used in the same way to wash the standstone sample, and finally compressed air is used to dry it. This design works well for medium-grained sandstone. For very fine-grained specimens, it seems preferable to drill a hole through the sample and to force the acid through the rock from the inside out. Bleached specimens subsequently can be treated

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by standard demagnetization techniques to remove secondary magnetic components within the remaining magnetic heavy mineral fraction.

Figure 2 shows a comparison of typical results obtained by the four different techniques: alternating-field, thermal, liquid nitrogen, and chemical demagnetization. Only the chemical demagnetization method produced consistent changes of NRM directions in this particular Lower Paleozoic rock sequence from an arid region. Intensities of NRM decreased considerably during the bleaching process.

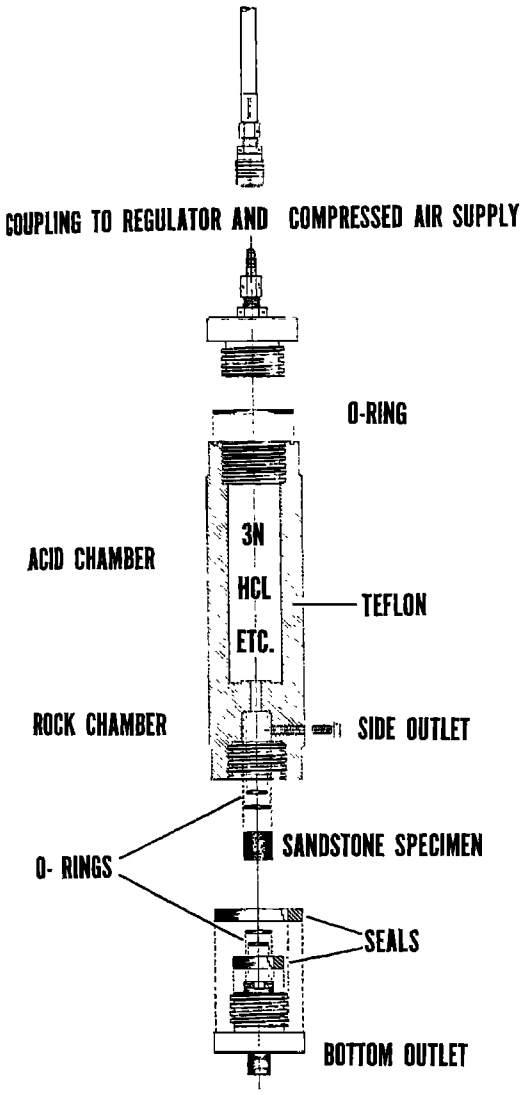


Fig. 1. Chemical demagnetization device.

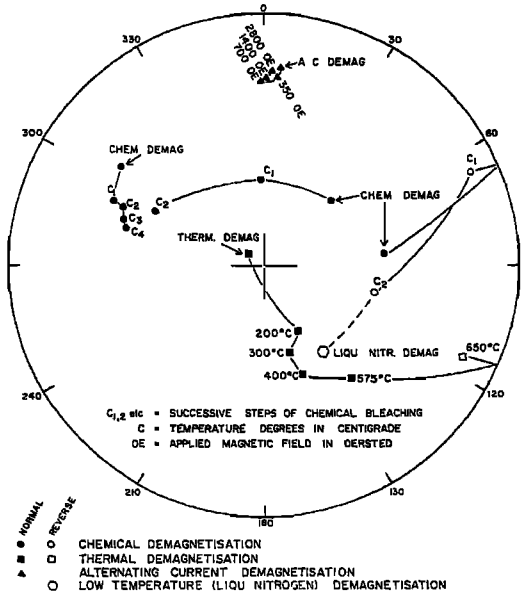


Fig. 2. Stability test examples obtained from four different demagnetization techniques.

The Cambro-Ordovician magnetic pole of the Arabian shield derived from these chemically demagnetized rocks is found to be close to the Azores (36.6°W, 36.7°N; N = 20, K = 20, $\alpha = 7.5$), which is consistent with other paleomagnetic results from Africa [*Briden*, 1968; *Brock*, 1967; *Graham et al.*, 1961] and evidence of Cambro-Ordovician glaciation in the Western Sahara [*Beuf et al.*, 1966].

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